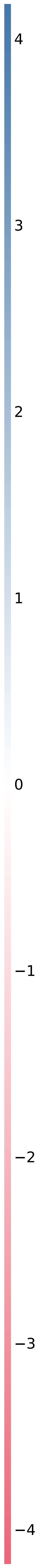
**Question:**
A 34-year-old woman presents to her OB/GYN with complaints of missing her last 3 periods as well as intermittent spontaneous milky-white nipple discharge bilaterally for the past 3 months. Vital signs are stable and within normal limits. Neurologic examination is without abnormality, including normal visual fields. Serology and MRI of the brain are ordered, with results pending. Which of the following sets of laboratory results would be expected in this patient?
Answer Choices:
A. Decreased prolactin, decreased FSH, decreased LH
B. Decreased prolactin, increased FSH, increased LH
C. Increased prolactin, decreased FSH, increased LH
D. Increased prolactin, decreased FSH, decreased LH




**Reasoning + Answer:**

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### Finding Reason ing Paths :
1 . ** P rol act in oma ( pit uit ary tumor )** : A prol act in -se cre ting tumor could lead to high prol act in levels , which may suppress the hypoth al amic -p it uit ary axis , causing decreased levels of F SH and LH .
2 . ** H yp oth al amic dysfunction **: Conditions affecting the hypoth alam us , such as rare diseases or structural abnormalities , could disrupt the release of releasing hormones , leading to decreased F SH and LH production .
3 . ** Pol yc yst ic ov ary syndrome ( PC OS )** : This condition can cause ovarian dysfunction , leading to decreased F SH and LH secretion .
4 . ** Med ic ations ( e .g ., met oc lo pr amide or dom per id one )** : These drugs can increase prol act in levels , potentially causing decreased F SH and LH .
5 . ** Secondary amen orr hea **: Horm onal im balances could be secondary to a pit uit ary or hypoth al amic cause , leading to decreased F SH and LH .

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### Reason ing Process :
1 . ** P rol act in oma ( pit uit ary tumor )** :
- P rol act in omas are the most common type of pit uit ary aden oma , and they are often micro aden omas .
- A prol act in -se cre ting tumor can cause hyper pro l act in emia , which suppress es gon ad ot rop in -re leasing hormone ( G n RH ) release from the hypoth alam us .
- This suppression leads to decreased secretion of F SH and LH from the anterior pit uit ary .
- The milk production and abnormal nipple discharge are consistent with this condition , as hyper pro l act in emia can lead to gal actor r hea and ovarian dysfunction .

2 . ** H yp oth al amic dysfunction ** :
- Conditions affecting the hypoth alam us , such as genetic synd rom es ( e .g ., McC une -Al bright syndrome ), can disrupt the release of Gn RH .
- This would result in decreased F SH and LH secretion , potentially leading to amen orr hea and other hormonal im balances .

3 . ** Pol yc yst ic ov ary syndrome ( PC OS )** :
- While PC OS is a common cause of hyper and rogen emia and infertility , it is less likely to be the cause in this case , as the patient has missed her last three periods and has not reported symptoms like acne or h irs ut ism .

4 . ** Med ic ations ( e .g ., met oc lo pr amide or dom per id one )** :
- These drugs are known to increase prol act in levels by binding to dopamine receptors in the pit uit ary gland .
- Elevated prol act in levels can suppress Gn RH release and lead to decreased F SH and LH production .

5 . ** Secondary amen orr hea ** :
- This condition refers to the absence of menstr uation in the absence of pregnancy , and it could be secondary to a pit uit ary or hypoth al amic cause .
- The combination of decreased F SH and LH is a hallmark of hyp og on ad otropic hyp og on ad ism , which could be caused by a pit uit ary aden oma or hypoth al amic dysfunction .

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### Conclusion :
The most likely cause of the decreased F SH and LH , along with the absence of m enses and milk production , is ** hyper pro l act in emia due to a prol act in -se cre ting pit uit ary aden oma ( pro l act in oma )** . This condition explains the gal actor r hea , amen orr hea , and decreased gon ad ot rop in levels .
</think >
< answer >
Increased prol act in , decreased F SH , decreased LH
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